Optimization involving a single variable

Q. The sum of 2 +ve numbers is 16.
What is the smallest possible value of the sum of their squares?

→ $x + y = 16$

$y = 16 - x$

$S = x^2 + y^2 = x^2 + (16-x)^2$

$S' = 2x - 2(16-x) = 4x - 32$ $\quad 4x - 32 = 0$

$S'' = 4$ $\quad x = 8$

$S'' > 0$ for $x = 8$

∴ The minimum value is at $x = 8$

$S(8) = 8^2 + (16-8)^2 = 128.$

GATE 2023 — Mechanical Engineering (ME)

Q.48	The smallest perimeter that a rectangle with area of 4 square units can have is _____ units. *(Answer in integer)*

→ Area = 4

$lb = 4$

$b = \frac{4}{l}$

$P = (l+b)2$

$P = \left(l + \frac{4}{l}\right)2 = 2l + \frac{8}{l}$

$P' = 2 - \frac{8}{l^2} \rightarrow 2 - \frac{8}{l^2} = 0 \Rightarrow \frac{8}{l^2} = 2 \Rightarrow l^2 = 4 \Rightarrow l = \pm 2$

$P'' = \frac{16}{l^3}$ at $l = 2$, $P'' = \frac{16}{8} = 2 > 0$

min. value of perimeter $= 2(l+b)$

$= 2\left(2 + \frac{4}{2}\right) = 8$ units

2012 MECHANICAL ENGINEERING - ME

Q.63 A political party orders an arch for the entrance to the ground in which the annual convention is being held. The profile of the arch follows the equation $y = 2x - 0.1x^2$ where y is the height of the arch in meters. The maximum possible height of the arch is

(A) 8 meters (B) 10 meters (C) 12 meters (D) 14 meters

→ $y = 2x - 0.1x^2$

$y' = 2 - 0.2x \rightarrow x = \frac{2}{0.2} = 10$

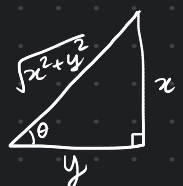
$y'' = -0.2 < 0$

$y(10) = 2(10) - 0.1(100) = 20 - 10 = 10$

Q.29 For a right angled triangle, if the sum of the lengths of the hypotenuse and a side is kept constant, in order to have maximum area of the triangle, the angle between the hypotenuse and the side is

(A) 12° (B) 36° (C) 60° (D) 45°

→



$x + \sqrt{x^2+y^2} = k$

$x^2 + y^2 = (k-x)^2 = x^2 + k^2 - 2kx$

$y^2 = k^2 - 2kx$

$2kx = k^2 - y^2$

$x = \frac{k^2 - y^2}{2k}$

$A = \frac{1}{2} \times 2xy$

$A = \frac{1}{2} \times \frac{k^2 - y^2}{2k} \cdot y = \frac{y}{4k}(k^2 - y^2)$

$A' = \frac{y}{4k}(-2y) + (k^2 - y^2)\frac{1}{4k} = 0$

$-\frac{2y^2}{4} + \frac{k^2}{4} - \frac{y^2}{4} = 0$

$k^2 - 3y^2 = 0$

$y^2 = \frac{k^2}{3}$

$y = \frac{k}{\sqrt{3}}$

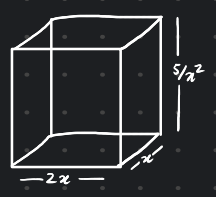
$x = \frac{2k^2}{3} \times \frac{1}{2k} = \frac{k}{3}$

$\tan\theta = \frac{x}{y} = \frac{k}{3} \times \frac{\sqrt{3}}{k} = \frac{1}{\sqrt{3}} \quad \theta = 30°$



angle could be either one. 30° or 60°.

Q. A rectangular container with an open top needs to have a volume of 10 cubic meters. The length of the base is twice its width. Material for the base costs $10 per sq m. Material for the sides costs $6 per sq m. Find the cost of the material for the cheapest container.



$vol^m = 10$

$2x \cdot x \cdot h = 10$

$h = \frac{5}{x^2}$

$cost = \left(\text{area of base}\right)(10) + 2(\text{area of 1 side})(6) + 2(\text{area of other side})(6)$

$c = (2x)(x)(10) + 2(x)\left(\frac{5}{x^2}\right)(6) + 2(2x)\left(\frac{5}{x^2}\right)(6)$

$c = 20x^2 + \frac{60}{x} + \frac{120}{x} = 20x^2 + \frac{180}{x}$

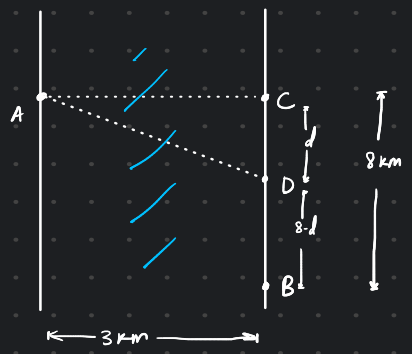
$c' = 40x - \frac{180}{x^2} = 0$

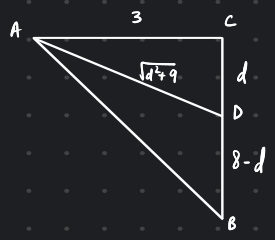
$40x = \frac{180}{x^2}$

$x^3 = \frac{180}{40} = \frac{9}{2} = 4.5 \quad \Rightarrow x = 1.6509$

$c = 163.5408$

Q. A man launches his boat from point A on a bank of straight river, 3 km wide and wants to reach a point B, 8 km downstream on the opposite bank. He could row his boat directly across the river to point C and then run to B, or he could row directly to B, or he could row to some point D between C and B and then run to B. If he can row 6 km/hr and run 8 km/hr, where should he land to reach B as soon as possible? We assume that the speed of water is negligible compared to the speed at which the man rows.





Total time $t = \frac{\sqrt{d^2+9}}{6} + \frac{8-d}{8}$ → minimize problem.

$t(d) = \frac{\sqrt{d^2+9}}{6} + \frac{8-d}{8}$

$t' = \frac{1}{6} \times \frac{1}{2\sqrt{d^2+9}} \cdot 2d - \frac{1}{8} = 0$

$\frac{d}{3\sqrt{d^2+9}} = \frac{1}{4}$

$16d^2 = 9(d^2+9)$

$7d^2 = 81$

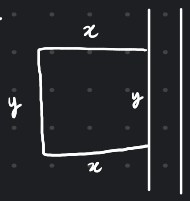
$d^2 = \frac{81}{7}$

$d = \frac{9}{\sqrt{7}} = 3.4016$

Should land 3.4 kms away from C.

Q. A farmer has 2400 ft of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field with maximum area?

→



$2x+y=2400$

$y=2400-2x$

Area $= xy$

$A = x(2400-2x)$

$= 2400x - 2x^2$

$A' = 2400 - 4x = 0$

$x = 600$

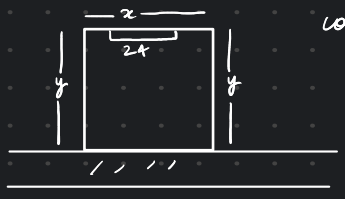
$y = 2400-1200 = 1200$

$x, y = 600, 1200$

$A'' = -4 < 0 \quad \therefore$ maxima.

Q. A rectangular lot is bounded at the back by a river. No fence is needed along the river. There's a 24 ft opening at the front. If the fence along the front costs \$1.50 per ft, along the sides \$1 per ft, find the dimensions of the largest lot which can thus be fenced in for \$300.

→



$\text{cost} = (x-24)(1.50) + (2y)(1) = 300$

$2y + \frac{3}{2}x - \frac{24 \cdot 3}{2} = 300$

$4y + 3x - 72 = 600$

$3x + 4y = 528$

$x = \frac{528-4y}{3}$

$A = xy = \left(\frac{528-4y}{3}\right)y = \frac{528y}{3} - \frac{4}{3}y^2$

$A' = \frac{528}{3} - \frac{8}{3}y = 0$

$y = \frac{528}{3} \times \frac{3}{8} = 66$

$x = \frac{528-4\cdot 66}{3} = 88$

mistake!

Q. A 1 ltr cylinder is made to hold oil. Find dimensions to minimize cost of metal used.

→ $\pi r^2 h = 1000$ cc

$h = \frac{1000}{\pi r^2}$

cost = that of surface area

surface area $= 2\pi rh + 2\pi r^2$

$A = 2\pi r \cdot \frac{1000}{\pi r^2} + 2\pi r^2$

$A = \frac{2000}{r} + 2\pi r^2$

$A' = -\frac{2000}{r^2} + 4\pi r = 0$

$A'' = 4\pi + \frac{4000}{r^3} > 0$ minima

$4\pi r = \frac{2000}{r^2}$ 500

$r^3 = \frac{500}{\pi}$

$r = 5.419 \qquad h = \frac{1000}{\pi r^2} = \frac{1000}{\pi \cdot (29.3683)} = 10.8385$

Q. cylinder. No top. 100 sq. inches. max. volume.

→ $2\pi rh + \pi r^2 = 100$

$h = \frac{100 - \pi r^2}{2\pi r}$

$\text{vol}^m = \pi r^2 h$

$= \pi r^2 \left(\frac{100-\pi r^2}{2\pi r}\right)$

$V = \frac{r(100-\pi r^2)}{2}$

$V = 50r - \frac{\pi r^3}{2}$

$V' = 50 - \frac{3\pi}{2} r^2$

$r = \sqrt{\frac{50 \times 2}{3\pi}} = \sqrt{\frac{100}{3\pi}} = 3.2573$

$h = \frac{100 - \pi \cdot \frac{100}{3\pi}}{2\pi \sqrt{\frac{100}{3\pi}}}$

$= \frac{2}{3} \cdot \frac{100}{\pi} \cdot \frac{\sqrt{3}\sqrt{\pi}}{\sqrt{100}}$

$= \frac{\sqrt{100}}{\sqrt{3\pi}} = \sqrt{\frac{100}{3\pi}}$

Q. 120 ft. material to enclose 2 rect. pens. Find x and y to max. area.

→



$2x+3y=120$

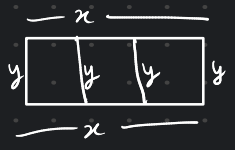
$x=\frac{120-3y}{2}$

$A=(y)\left(\frac{120-3y}{2}\right)$

(30, 20)

Q. Rect. field, fixed area. To be enclosed and divided into 3 lots by parallels to one of the sides. Relative dimensions of field to make the amount of fencing minimum?

→



$xy=k$

$y=\frac{k}{x}$

Fencing

$F=4y+2x$

$F=\frac{4k}{x}+2x$

$F'=-\frac{4k}{x^2}+2=0$

$2=\frac{4k}{x^2}$

$x^2=\frac{4k}{2}$

$x=\sqrt{2}\sqrt{k}$

$y=\frac{k}{x}=\frac{k}{\sqrt{2}\sqrt{k}}=\frac{\sqrt{k}}{\sqrt{2}}$

$\frac{x}{y}=\frac{\sqrt{2}\sqrt{k}}{\sqrt{k}}\sqrt{2}=\frac{2}{1}$

Q. Window, rect with ½ circle. Only 1m of frame material. Frame runs around outside of the window and betn rectangular and semi-circular region. Diameter of semicircle to maximize area of window?

→



framing material $=2x+2y+\pi\left(\frac{x}{2}\right)=1m$

$y=\frac{1-2x-\frac{\pi}{2}x}{2}$

Area $=xy+\frac{1}{2}\pi\frac{x^2}{4}$

$A=x\left(\frac{1}{2}-x-\frac{\pi}{4}x\right)+\frac{\pi}{8}x^2$

$A=\frac{x}{2}-x^2-\frac{\pi}{4}x^2+\frac{\pi}{8}x^2$

$A=\frac{x}{2}-x^2-\frac{\pi}{8}x^2$

$A'=\frac{1}{2}-2x-\frac{\pi}{8}x=0$

$x=\frac{2}{\pi+8}$

Q. A cricket match is played at a stadium that holds 50,000 spectators with the ticket price $10. Average # of attendants was 27000. When the ticket prices were lowered to $8, the average rose to 33,000.
a. Find the demand function (ticket price) (assume linear).
b. Find the ticket price for the maximum revenue.

→ # of spectators = s price = p

(27, 10), (33, 8)

$\frac{y-8}{x-33}=\frac{8-10}{33-27}=\frac{-2}{6}=-\frac{1}{3}$

$3y-24=33-x$

$x+3y=57$

spectators (in thousands) $+3(\text{price})=57$

Revenue = (spectators) * (price)

$=(\text{price})(57-3\text{price})$

$R=p(57-3p)$

$R'=57-6p=0$

$6p=57$

$p=9.5$